4.5

Find a solution to the Schrödingerequation

$$\frac{\partial \psi}{\partial t} - i\frac{\partial^2 \psi}{\partial x^2} = 0, \quad t > 0, x \in \mathbb{R}$$

that satisfy the initial value

$$\psi(x, 0) = \sin^2(x), \quad x \in \mathbb{R}$$

Solution:

We will partial Fourier transform the differential equation, which results in

$$\int_{-\infty}^{\infty} \frac{\partial \psi}{\partial t} e^{-ix\xi} \, \mathrm{d}x + i\xi^2 \int_{-\infty}^{\infty} \psi e^{-ix\xi} \, \mathrm{d}x = 0$$

$$\downarrow$$

$$\frac{\partial \hat{\psi}}{\partial t}(\xi, t) + i\xi^2 \hat{\psi}(\xi, t) = 0$$

$$\downarrow$$

$$\hat{\psi}(\xi, t) = c(\xi) e^{-i\xi^2 t}$$

To determine $c(\xi)$, we set $t = 0$ and get

$$\hat{\psi}(\xi, 0) = c(\xi) = \int_{-\infty}^{\infty} u(x, 0) e^{-ix\xi} \, \mathrm{d}x = \int_{-\infty}^{\infty} \sin^2(x) e^{-ix\xi} \, \mathrm{d}x =$$

$$= \int_{-\infty}^{\infty} \left(\frac{e^{ix} - e^{-ix}}{2i} \right)^2 e^{-ix\xi} \, \mathrm{d}x = -\frac{1}{4} \int_{-\infty}^{\infty} (e^{2ix} - 2 + e^{-2ix}) e^{-ix\xi} \, \mathrm{d}x$$

$$\downarrow$$

$$c(\xi) = -\frac{1}{4} \mathcal{F} \left[e^{2ix} - 2 + e^{-2ix} \right] = -\frac{1}{4} [2\pi\delta(\xi - 2) - 4\pi\delta(\xi) + 2\pi\delta(\xi + 2)] =$$

$$\rightarrow \boxed{c(\xi) = \frac{\pi}{2} [2\delta(\xi) - \delta(\xi - 2) - \delta(\xi + 2)]}$$

This gives our $\hat{\psi}(\xi, t)$

$$\hat{\psi}(\xi, t) = \frac{\pi}{2} [2\delta(\xi) - \delta(\xi - 2) - \delta(\xi + 2)] e^{-i\xi^2 t} =$$

$$= \pi\delta(\xi) - \frac{\pi}{2} e^{-4it} \delta(\xi - 2) - \frac{\pi}{2} e^{-4it} \delta(\xi + 2)$$

and our solution $\psi(x, t)$ is the inverse Fourier transform of $\hat{\psi}(\xi, t)$

$$\mathcal{F} \left[\frac{1}{2} - \frac{1}{4} e^{-4it} e^{2ix} - \frac{1}{4} e^{-4it} e^{-2ix} \right] = \pi\delta(\xi) - \frac{\pi}{2} e^{-4it} \delta(\xi - 2) - \frac{\pi}{2} e^{-4it} \delta(\xi + 2)$$

So our solution that satisfy the initial conditions is

$$\psi(x,t) = \frac{1}{2} - \frac{1}{4}e^{-4it}e^{2ix} - \frac{1}{4}e^{-4it}e^{-2ix} = \frac{1}{2}\left(1 - e^{-4it}\left[\frac{e^{2ix}+e^{-2ix}}{2}\right]\right)$$

$$\downarrow$$

$$\boxed{\psi(x,t) = \frac{1}{2}(1 - e^{-4it}\cos(2x))}$$

□